

Nodecraft.io - PPR Modeling

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Overview

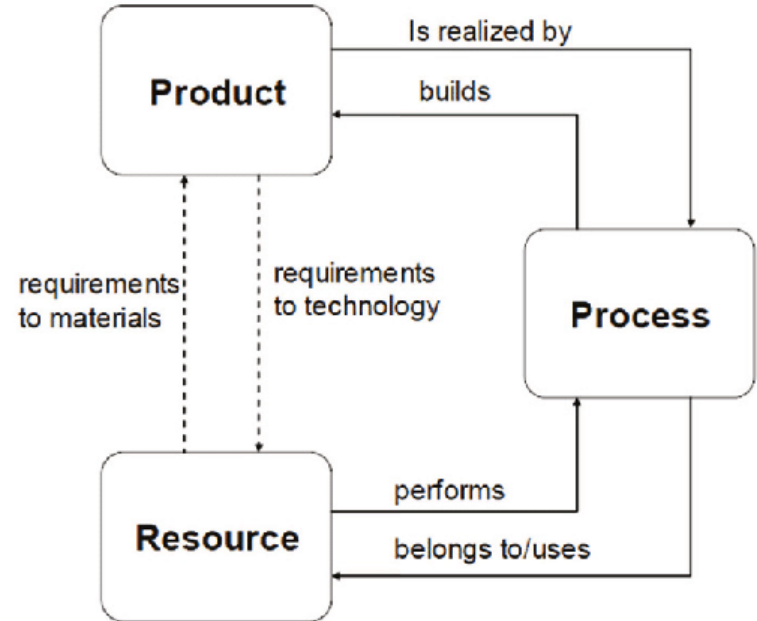
- What is PPR Model
- Nodecraft.io - Web Application
- Technology Stack
- System Architecture
- Database Schema
- UI UX Design
- Key Features
- Additional Features
- Development Methodology
- Challenges & Lessons Learned
- Future Work



PPR Model

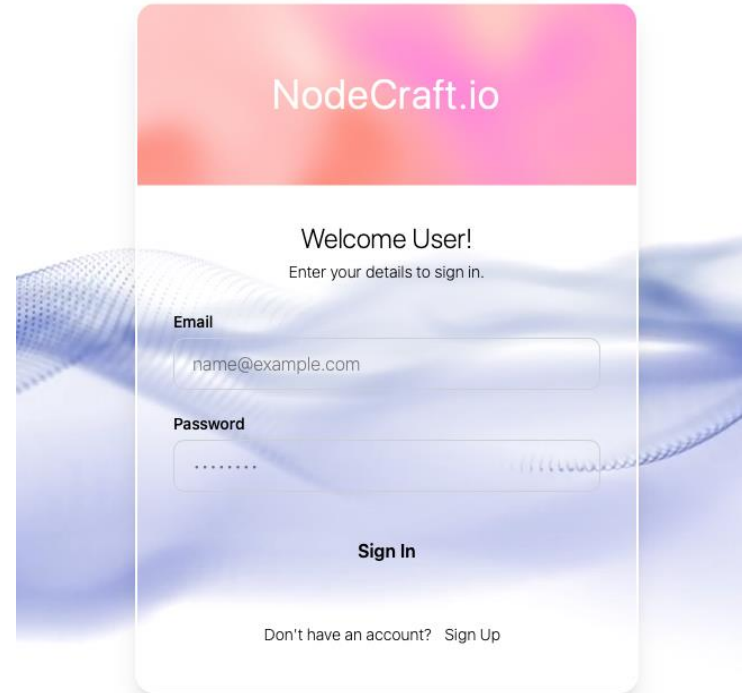
Product-Process-Resource Model is an integrated manufacturing framework that maps the essential interdependencies of production.

- **Product:** *What is being built?*
- **Process:** *How is it built?*
- **Resource:** *Who or What builds it?*



Nodecraft.io - Web Application

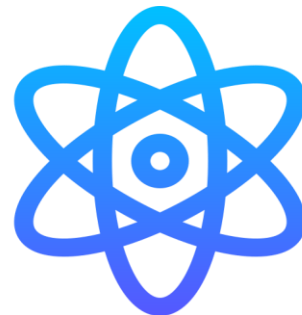
- Full-stack web application designed for visual process modeling and structured data management
- Bridges the gap between static engineering drawings and dynamic digital systems
- Provides "Infinite Canvas" that allows mapping of complex engineering systems



Technology Stack

Frontend & UI

- React.js, TypeScript
- React Flow
- Framer Motion , Shadcn
- Tailwind CSS



Backend & Data

- Google Firebase
- Cloud Firestore
- Firebase Auth



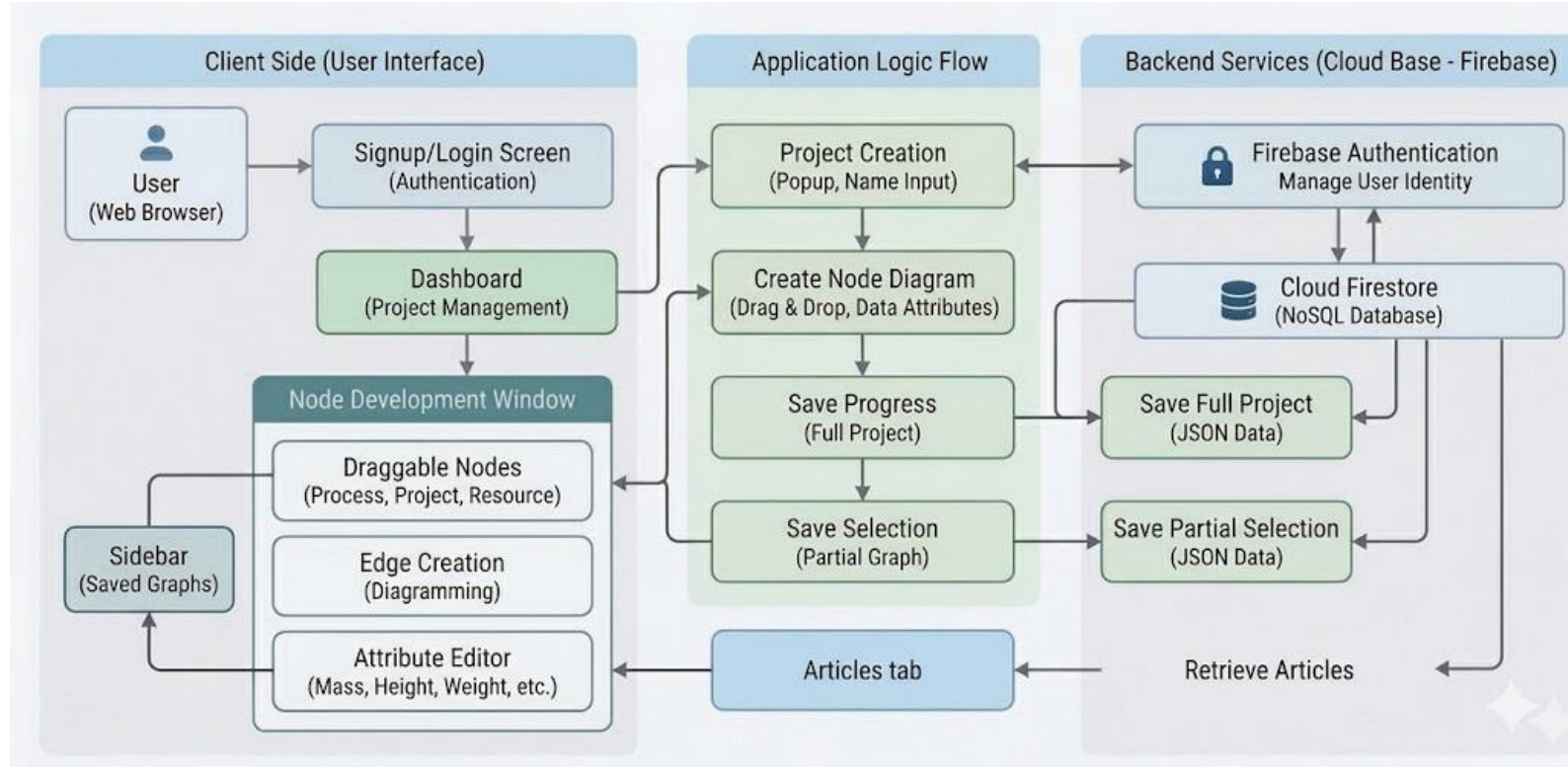
git

▲ Vercel

DevOps & Tools

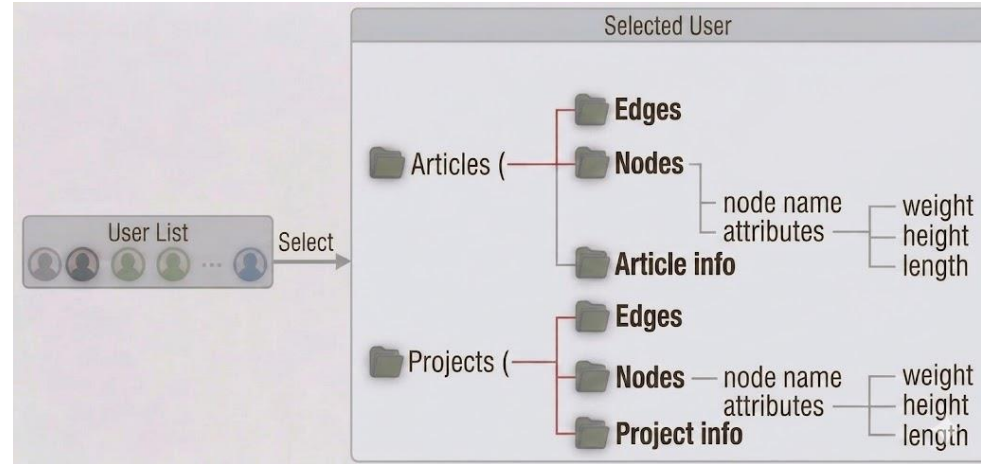
- Vercel
- GitHub
- Nanoid

System Architecture

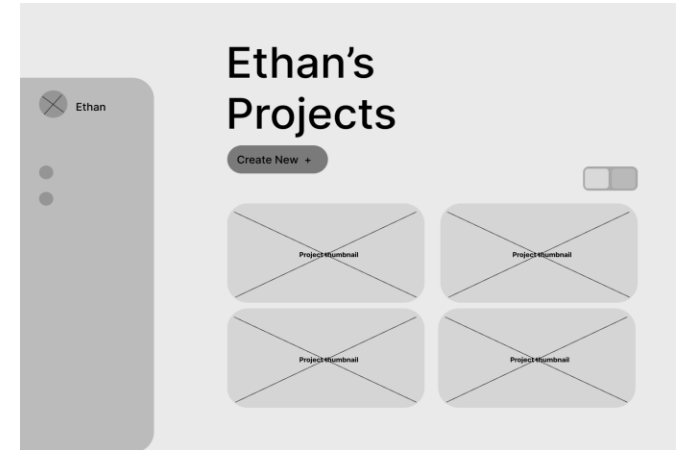
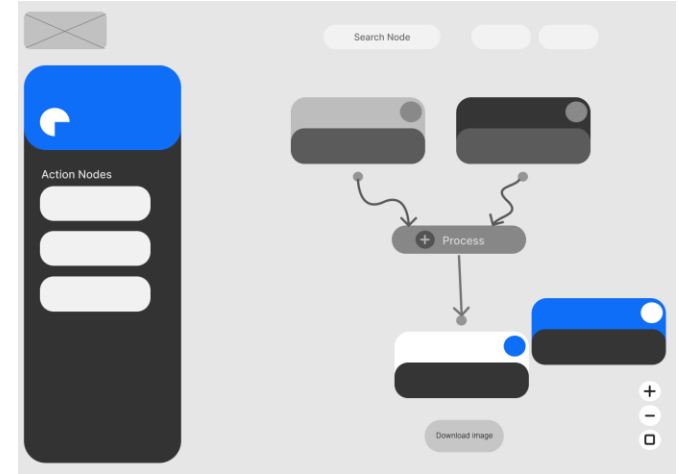


Database Schema

- Database consists of list of users
- Each user has separate sections for Articles, and Projects
- Project includes name, number of Nodes and Edges
- Node has a node name and attributes



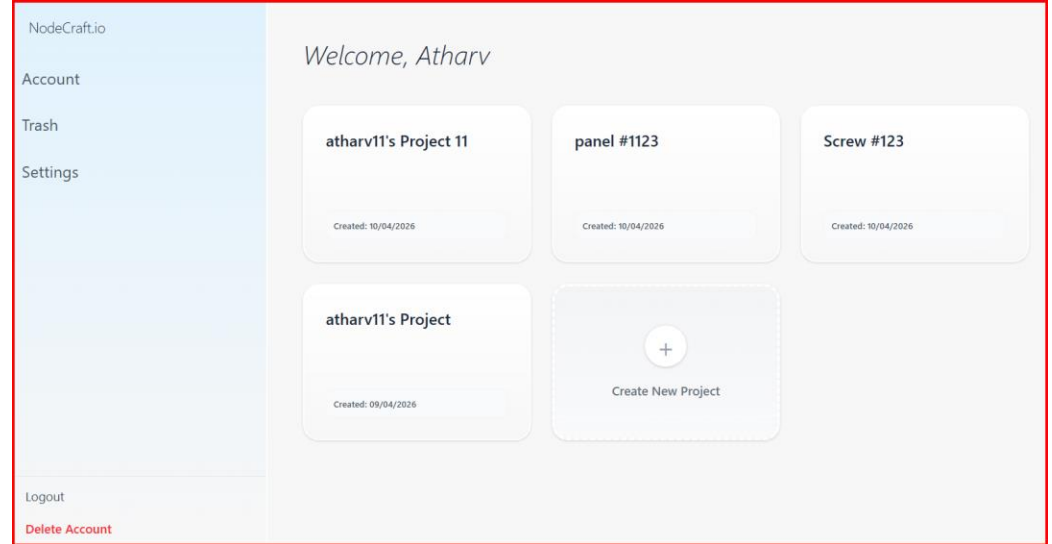
- Designed three core web pages - Login, Dashboard and Playground
- Created modern, low-fidelity and high-fidelity wireframes using Figma
- Added smooth animations to create a more engaging user experience



Key Features

User Dashboard

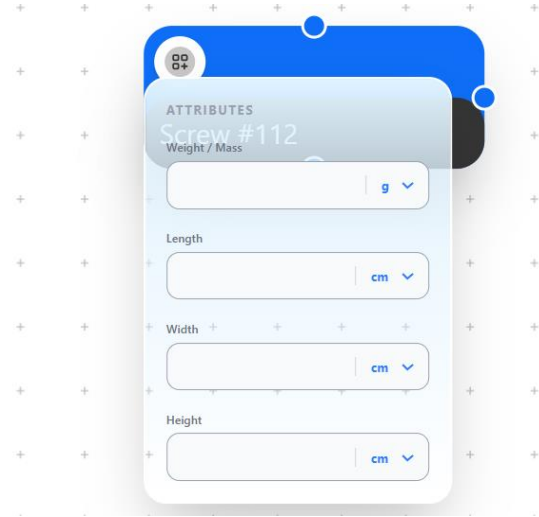
- A single space to view and manage all your work
- Create and store multiple independent projects without overlap



Key Features

Attributes

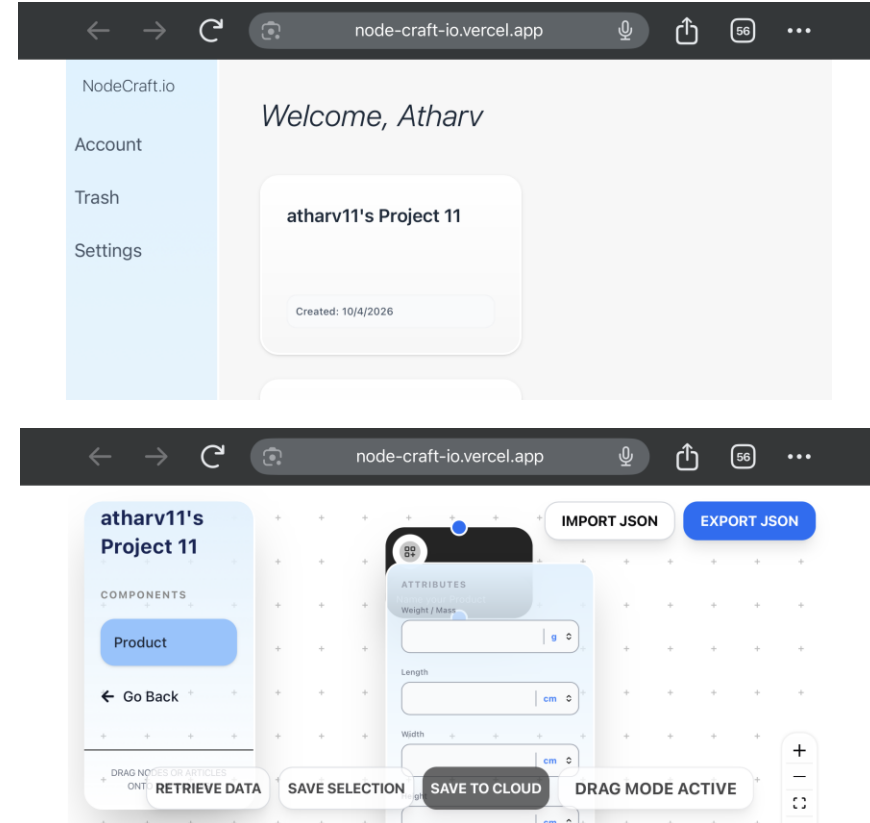
- Users can define node parameters, such as geometric dimensions and weight
- Toggle easily between different units for dimensions and weight



Key Features

Responsive Design

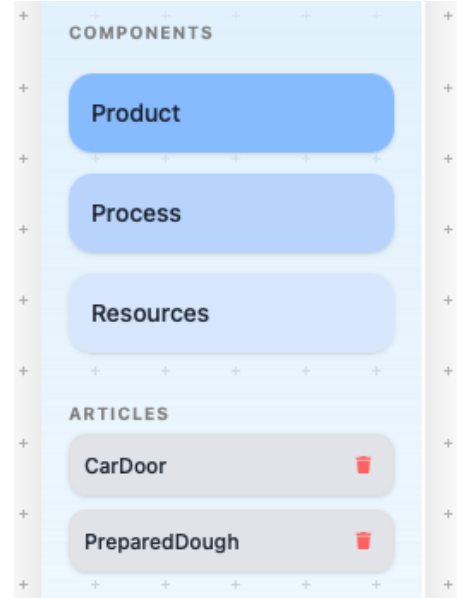
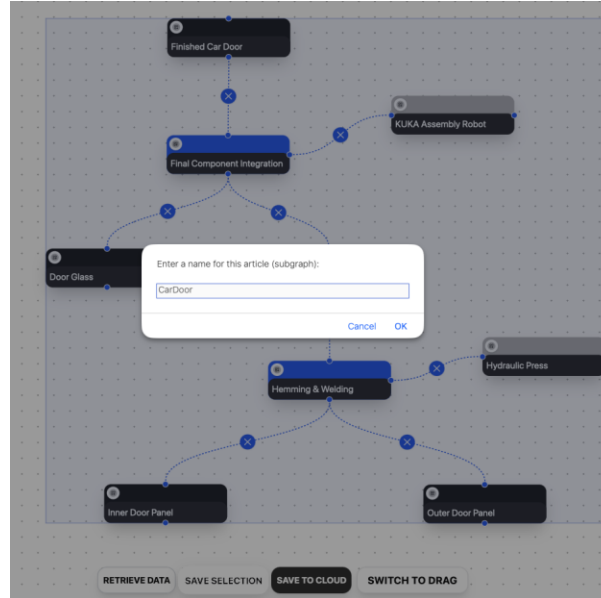
- User can access the website on most of the devices
- accessing this website on a desktop or tablet gives optimal experience



Key Features

Articles

- One of NodeCraft.io's most powerful features is the ability to save Subgraphs
- Select any part of a complex graph and save it as a standalone "Article"
- Re-import these "Articles" into other projects as building blocks



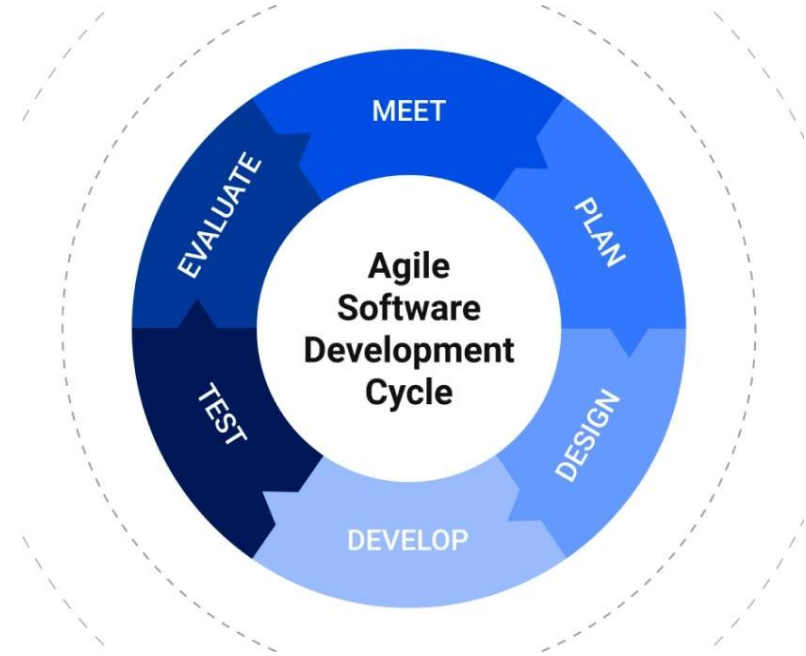
Additional Features

- **Import JSON and Export JSON:** Enables users to download their entire workflow as a lightweight JSON file
- **User Email Verification:** Implemented a mandatory verification layer to ensure only authenticated users use the platform



Built with Agility

- Meet & Plan: Defining the goal and features we needed to implement
- Design: Mapping the React Flow architecture
- Develop: Iterative sprints for the logic engine and UI
- Test & Evaluate: Verification and Validation of developed feature



User Feedback

Positive Feedback

- Significantly reduces modeling time through use of Articles
- JSON Import-Export enables seamless collaboration and project sharing
- UI feels modern and easy to use

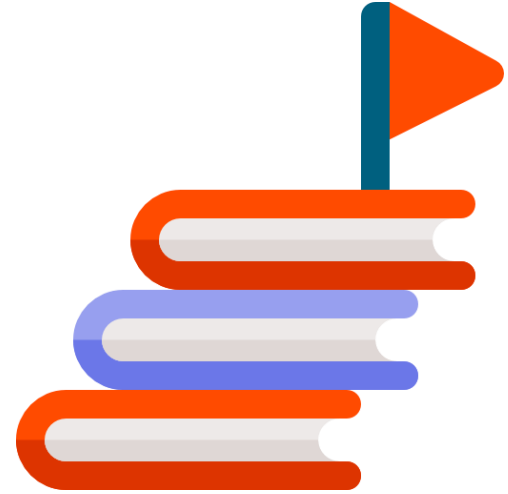
Suggestion Received

- Make the application iPad and tablet friendly



Challenges & Lessons Learned

- Switching from Realtime Database to Firestore Database
- The Articles feature blocked users from dragging and navigating to specific nodes
- Fixed design issues with the nodes to make them easier to use
- Used Nano id to get unique identifier and avoid the id conflict



Future Work

- As number of users increase database optimization will need more work
- Real time multi-user collaboration
- Mobile friendly design and gesture optimization
- Adding more user features on dashboard



Thank you for your attention!

Github Link: <https://github.com/atharv11/NodeCraft.io.git>

Web Application: <https://node-craft-io.vercel.app/>